UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS07 - Degree Distributions, Scale-Free Networks, and Robustness

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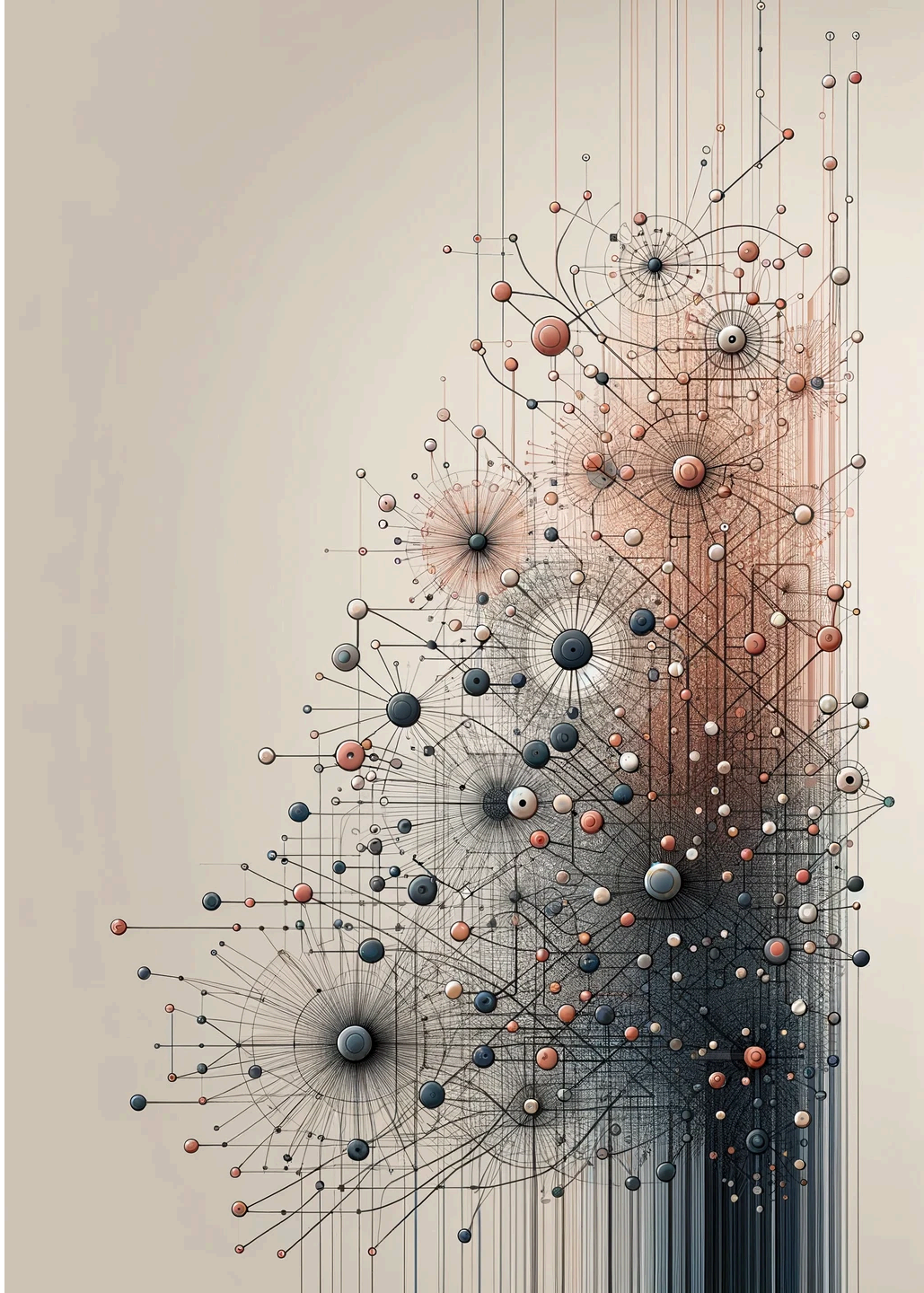


Outline

1. Degree distributions as statistical descriptions
2. Heavy tails, power laws, and scale-free claims
3. Heterogeneity and hubs
4. Consequences of hubs: sampling, paths, robustness

From NS06 to NS07

- In `NS06` we asked which nodes are important
- In `NS07` we move from individual nodes to **distributions of node properties**
- The main example is the **degree distribution**
- The same statistical language can later be applied to other measures

Degree distributions

For each degree value k, we can count how many nodes have that degree.

- **Count**: n_k = number of nodes with degree k
- **Frequency**: $f_k = \frac{n_k}{N}$
- **Probability**: in large networks, f_k approximates the degree distribution $p(k)$

Example:

- if 100 nodes out of 1,000 have degree 3, then $f_3 = 0.1$

Degree distribution as a PMF

For the **degree distribution**:

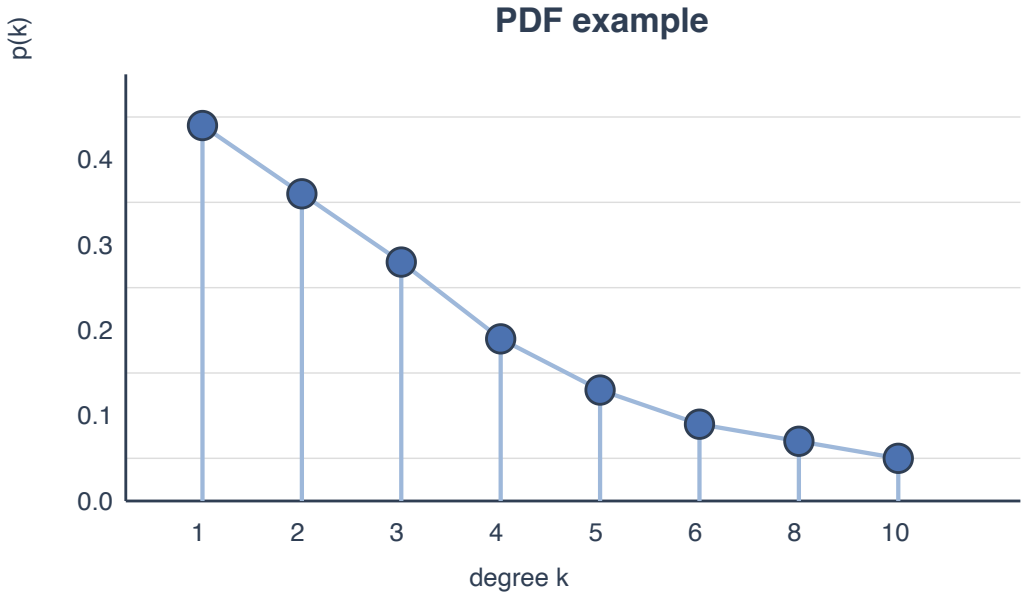
- $p(k)$ = probability that a randomly chosen node has degree k
- plot $p(k)$ against k

What it tells us:

- whether most nodes have small degree
- whether a few nodes act as hubs
- how broad the distribution is

Example: degree distribution

- For degree, the plot shows the probability mass at each exact integer value
- Use it when the question is:
  - how frequent is degree k?
  - where is the mass concentrated?



Cumulative Distribution Function (CDF)

The **CDF** gives the probability that the degree is at most x:

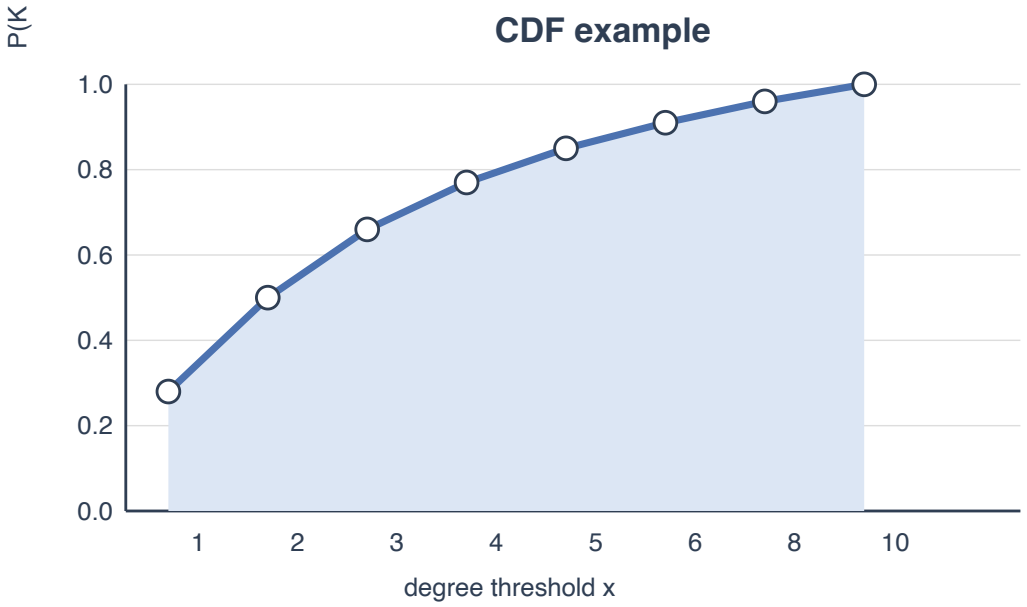
$$P(X \leq x) = \sum_{v \leq x} p(v)$$

It accumulates probability from small values to large values.

- starts near 0
- increases monotonically
- ends at 1

Example: CDF

- The CDF accumulates probability from the left
- Use it when the question is:
  - what fraction of nodes has degree at most x?
  - how quickly does probability accumulate?
- The plotted values are evaluated at integer degree thresholds



Complementary Cumulative Distribution Function (CCDF)

The **CCDF** gives the probability that the degree is larger than x:

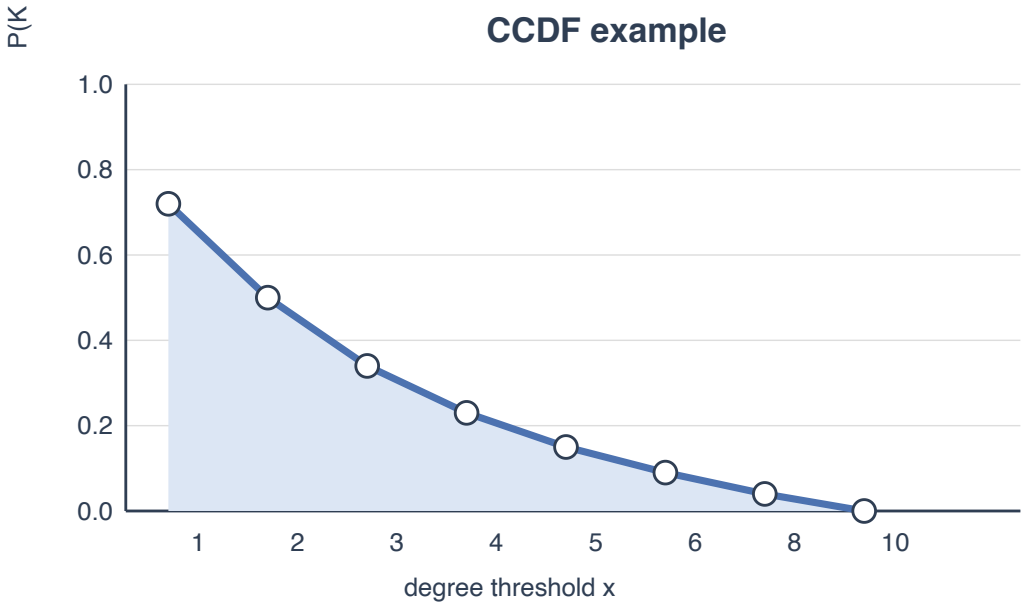
$$P(X > x) = \sum_{v>x} p(v)$$

Why it is useful:

- it emphasizes the tail
- it is smoother than the raw PDF
- it is often the clearest way to inspect broad or heavy-tailed distributions

Example: CCDF

- The CCDF focuses on the remaining probability in the tail
- Use it when the question is:
  - how many nodes have degree larger than x?
  - how important is the tail of the distribution?
- The plotted values are evaluated at integer degree thresholds



How to read these distributions

- **PDF**:
  read height at degree k as the probability mass at that exact value
- **CDF**:
  choose a threshold x, then read the fraction of nodes with degree at most x
- **CCDF**:
  choose a threshold x, then read the fraction of nodes with degree larger than x

What to look for:

- where most of the mass is concentrated
- how broad the distribution is
- whether the tail decays quickly or slowly

Which one to use

- Use the **PDF** when you care about exact values or local density:
  - which degree is most common?
  - where is the mass concentrated?
- Use the **CDF** for threshold or percentile questions:
  - how many nodes have degree at most 5?
  - where does 80% of the probability accumulate?
- Use the **CCDF** to study the tail:
  - how many hubs are there?
  - how quickly does the tail decay?
  - is the distribution broad or heavy-tailed?

In practice:

- PDF = local
- CDF = cumulative
- CCDF = tail-focused

Which one to use in practice

For broad distributions:

- the **PDF** can be noisy
- the **CCDF** is often the clearest plot for the tail

Continuous variables: same ideas, different look

- For a **continuous** variable, the same ideas are used:
  - **PDF** describes local density
  - **CDF** accumulates probability from the left
  - **CCDF** shows how much probability remains in the tail
- The curves look smoother because the variable is not restricted to integer values
- For a continuous variable, the probability of one exact value is typically 0: probability is read from the **area over an interval**, not from the height at a single point

Discrete vs. continuous: what changes in practice

- **Discrete example**: degree
  - the PDF is a set of masses at integer values
  - the CDF and CCDF are evaluated at integer thresholds
- **Continuous example**: betweenness, closeness, weights, response times
  - the same three objects exist
  - the visual appearance is smoother
- **Empirical continuous data**:
  - the PDF is usually **estimated**
  - this is where **binning** matters: the shape of the histogram/PDF depends on the bins
  - kernel smoothing is another possible estimation approach
  - the CDF and CCDF can often be computed directly from sorted observations

What is binning?

For a continuous variable, two observations are rarely exactly equal.

To build an empirical PDF, we group nearby values into intervals called **bins**.

Example:

- `[0.0, 0.1)`
- `[0.1, 0.2)`
- `[0.2, 0.3)`

Then we:

1. count how many observations fall in each interval
2. normalize the counts
3. obtain an estimated PDF

Why binning matters

- If bins are **too wide**, the plot looks too smooth and hides structure
- If bins are **too narrow**, the plot looks noisy and unstable
- The same data can produce different-looking PDFs under different bin choices

Rule of thumb:

- for continuous empirical data, the **CDF/CCDF** are often more stable than the **PDF**
- if you show a continuous-data PDF, specify the **binning choice**

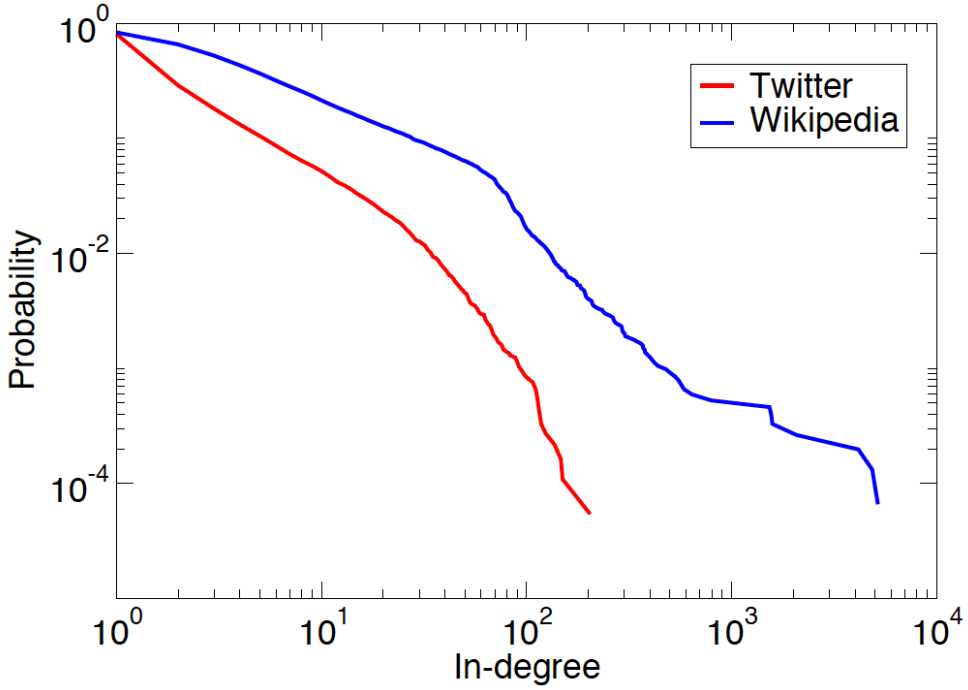
Logarithmic scale

When values span several orders of magnitude, linear axes hide the tail.

Use a **logarithmic scale** to compare small and large values in the same plot.

$$\log_{10} 10 = 1$$
$$\log_{10} 1,000 = \log_{10} 10^3 = 3$$
$$\log_{10} 1,000,000 = \log_{10} 10^6 = 6$$

A broad degree distribution



Heavy-tailed distributions contain many small values and a non-negligible number of very large ones.

Heavy-tailed, power-law, scale-free

- A **heavy-tailed** degree distribution decays slowly and can produce hubs
- A **power law** is a specific functional form:

$$P(k) \sim k^{-\gamma}$$

- A network is often called **scale-free** when its degree distribution is modeled as a power law and hubs dominate the structure
- If $p(k) \sim k^{-\gamma}$, then the CCDF satisfies:

$$P(K > k) \sim k^{-(\gamma-1)}$$

These are related ideas, but they are not interchangeable.

Caution on log-log plots

- A visually straight line on a log-log plot is **not proof** of a power law
- Log-log plots are useful diagnostics, not sufficient evidence
- Better wording:
  - **consistent with** a power law
  - **suggestive of** heavy-tailed structure
  - not **proves that the network is scale-free**

Recognizing broad tails

- **Exponential tail**: fast decay, no dominant hubs, often clearer on a semi-log plot
- **Heavy tail**: broad distribution with rare large values
- **Power law**: a stronger claim than “broad” or “heavy-tailed”
- The CCDF is often the cleanest visual summary of the tail

Typical broad-tailed networks

- Historically, broad degree distributions have been reported in:
  - the WWW
  - citation networks
  - Internet AS graphs
  - some biological interaction networks
- In such networks, hubs can affect:
  - sampling
  - path lengths
  - robustness

Heterogeneity Parameter κ

The **heterogeneity parameter** measures how broad the degree distribution is.

$$\langle k^2 \rangle = \frac{k_1^2 + k_2^2 + \cdots + k_{N-1}^2 + k_N^2}{N} = \frac{\sum_i k_i^2}{N}$$

$$\langle k \rangle = \frac{\sum_i k_i}{N} = \frac{2L}{N}$$

$$\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle^2}$$

Interpreting κ

If most nodes have similar degree, say k_0:

$$\langle k \rangle \approx k_0, \ \langle k^2 \rangle \approx k_0^2 \implies \kappa \approx 1$$

If the degree distribution is broad and hubs are present:

$$\kappa \gg 1$$

- small κ: no strong hubs
- large κ: hubs strongly affect network structure
- for directed graphs, study **in-degree** and **out-degree** separately

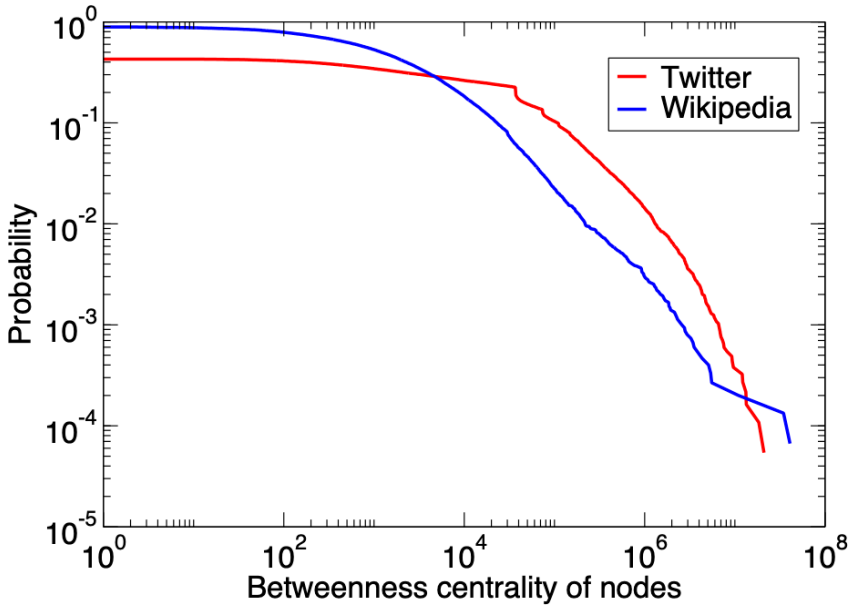
Degree distributions of real-world networks

Network	Nodes (N)	Links (L)	Average degree ($\langle k \rangle$)	Maximum degree (k_{max})	Heterogeneity parameter (κ)
Facebook Northwestern Univ.	10,567	488,337	92.4	2,105	1.8
IMDB movies and stars	563,443	921,160	3.3	800	5.4
IMDB co-stars	252,999	1,015,187	8.0	456	4.6
Twitter US politics	18,470	48,365	2.6	204	8.3
Enron Email	36,692	367,662	10.0	1,383	14.0
Wikipedia math	15,220	194,103	12.8	5,171	38.2
Internet routers	190,914	607,610	6.4	1,071	6.0
US air transportation	546	2,781	10.2	153	5.3
World air transportation	3,179	18,617	11.7	246	5.5
Yeast protein interactions	1,870	2,277	2.4	56	2.7
C. elegans brain	297	2,345	7.9	134	2.7
Everglades ecological food web	69	916	13.3	63	2.2

Other distributions

- The same statistical language can be applied to other quantities:
  - betweenness
  - closeness
  - eigenvector centrality
- Degree remains the main example because it is discrete, directly interpretable, and central to network models.
- Other measures are often continuous:
  - they usually require **binning**
  - **logarithmic binning** is often useful for skewed distributions

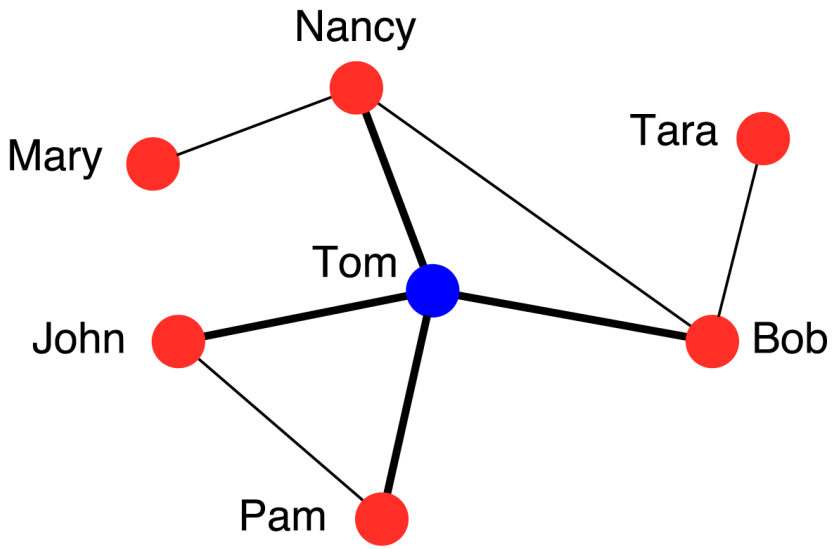
Example: betweenness distribution



Consequences of hubs

1. sampling becomes biased toward high-degree nodes
2. shortest paths become shorter
3. random failures and targeted attacks behave differently

Friendship paradox



- Sampling a node uniformly gives every node the same chance
- Sampling by following an edge favors high-degree nodes
- Hubs are therefore overrepresented in “friends of random people”

Friendship paradox

There are two different ways to sample degree:

- **Uniform node sampling**:
  every node has the same probability

$$\mu_f = \frac{\sum_i k_i}{N}$$

- This is the ordinary average degree of the network

Friendship paradox

If we follow a random edge, node i is reached with probability

$$P(i) = \frac{k_i}{\sum_j k_j}$$

Therefore the average degree of a friend is

$$\mu_{fof} = \sum_i P(i) k_i = \frac{\sum_i k_i^2}{\sum_i k_i}$$

- High-degree nodes contribute more because they are incident to more edges

Friendship paradox: formal consequence

By following edges, high-degree nodes are sampled more often:

$$\mu_{fof} = \frac{\sum_i k_i^2}{\sum_i k_i} \geq \frac{\sum_i k_i}{N} = \mu_f$$

- the paradox is a **sampling effect**
- the more heterogeneous the degree distribution, the stronger the gap between μ_{fof} and μ_f

Implications of the friendship paradox

The friendship paradox has practical applications in:

- **Epidemic monitoring**: Monitoring friends of random people detects outbreaks earlier
- **Social influence**: People tend to feel less popular than they actually are
- **Marketing**: Friends of random users may be better targets for campaigns
- **Vaccination strategies**: Immunizing friends of random individuals can be more effective than random vaccination

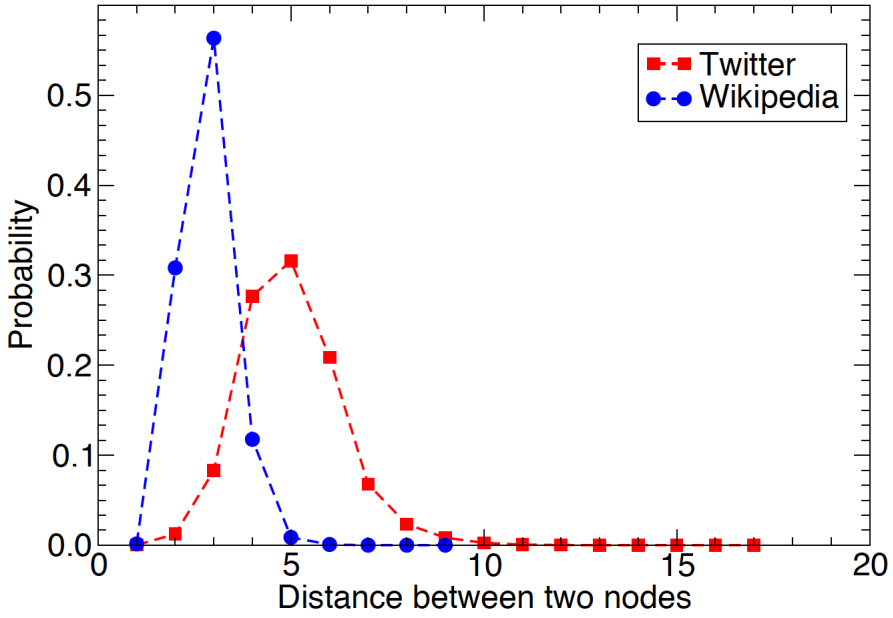
The key lesson is structural:

- hubs bias what we observe
- broad degree distributions amplify that bias

Ultra-small worlds

- Hubs act as shortcuts in the network
- Many shortest paths pass through a small number of high-degree nodes
- In transportation networks, two small airports may only be connected through a hub

Broad degree distributions can therefore produce **ultra-short paths**.



From hubs to failure modes

The same hubs that shorten paths also determine vulnerability

Robustness

A network is **robust** if failures do not destroy its global function.

For a connected network, the main structural question is:

- does the network remain connected when nodes are removed?

Operational measure:

- remove nodes
- track the relative size S of the largest connected component
- if S stays large, the network remains functional

Robustness

- **Random failures** mostly remove low-degree nodes, because they are the majority
- **Targeted attacks** remove hubs, which deletes many shortcuts and disconnects the network quickly

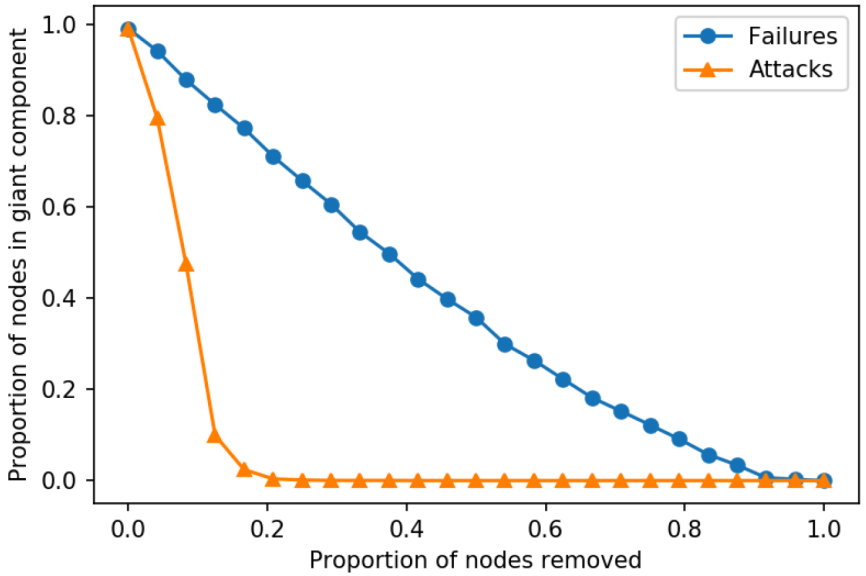
Formally, the two procedures sample the degree distribution differently:

- random failure samples from the bulk of $p(k)$, where low-degree nodes are concentrated
- attack targets the tail of $p(k)$, where the hubs concentrate many incident edges

This is why heterogeneity matters:

- broad degree distributions protect against random failure
- the same hubs create vulnerability under attack

Robustness



Typical outcome in heterogeneous networks:

1. **Robust against random failures**
2. **Fragile against targeted attacks**

The reason is structural: hubs are rare, but they carry a disproportionate share of connectivity.

Synthesis

- Degree distributions describe large networks statistically
- Heavy tails create hubs
- Hubs bias sampling and generate the friendship paradox
- Hubs shorten paths and can create ultra-small worlds
- Hubs also make networks fragile under targeted attack

A natural next question is which network models generate broad degree distributions and hub structure.

Reading material

References

[ns1] **Chapter 3**

[paper] Clauset, Shalizi, Newman. **Power-law distributions in empirical data**, SIAM Review, 2009

Q&A